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Roll NO: 21

Subject: Data Engineering

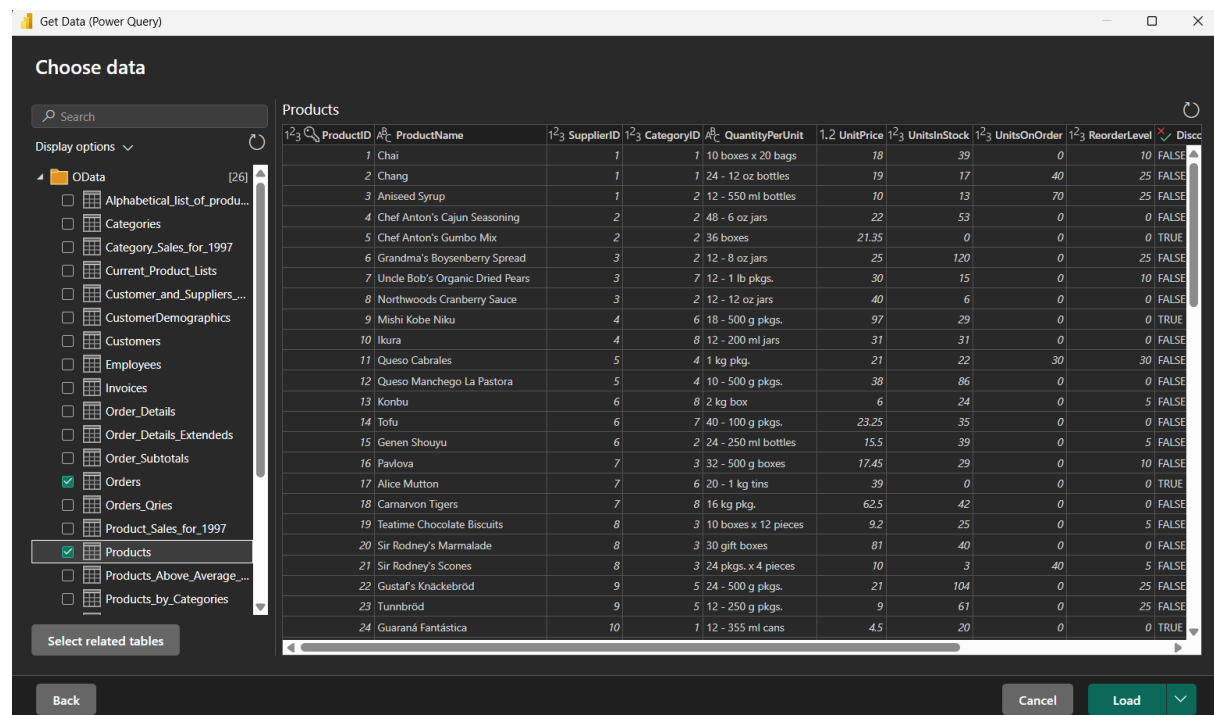
Practical No. – 2

Aim:

To perform the ETL process in Power BI by extracting, transforming, and loading data to prepare it for analysis.

Step 1- Import Excel Data into Power BI

The Excel workbook containing the Products, Orders, and Order_Details tables was imported into Power BI for further data transformation and analysis.



Step 2- Open Power Query Editor

Power Query Editor was opened to perform the required data transformation operations on the imported tables.

Power Query Editor

Home Transform Add column View Help

Close & Apply, Get data, Enter data, Manage parameters, Refresh, Properties, Advanced editor, Export query results, Choose columns, Remove columns, Keep rows, Remove rows, Filter rows, Transform, Combine, Export template

Queries [2]: Orders, Products

Table.ReorderColumns(#"Added custom", {"OrderID", "CustomerID", "EmployeeID", "OrderDate", "RequiredDate", "ShippedDate", "ShipVia"})

OrderID	CustomerID	EmployeeID	OrderDate	RequiredDate	ShippedDate	ShipVia
1	10248	VINET	5	7/4/1996, 12:00:00 AM	8/1/1996, 12:00:00 AM	7/16/1996, 12:00:00 AM
2	10248	VINET	5	7/4/1996, 12:00:00 AM	8/1/1996, 12:00:00 AM	7/16/1996, 12:00:00 AM
3	10248	VINET	5	7/4/1996, 12:00:00 AM	8/1/1996, 12:00:00 AM	7/16/1996, 12:00:00 AM
4	10249	TOMSP	6	7/5/1996, 12:00:00 AM	8/16/1996, 12:00:00 AM	7/10/1996, 12:00:00 AM
5	10249	TOMSP	6	7/5/1996, 12:00:00 AM	8/16/1996, 12:00:00 AM	7/10/1996, 12:00:00 AM
6	10250	HANAR	4	7/8/1996, 12:00:00 AM	8/5/1996, 12:00:00 AM	7/12/1996, 12:00:00 AM
7	10250	HANAR	4	7/8/1996, 12:00:00 AM	8/5/1996, 12:00:00 AM	7/12/1996, 12:00:00 AM
8	10250	HANAR	4	7/8/1996, 12:00:00 AM	8/5/1996, 12:00:00 AM	7/12/1996, 12:00:00 AM
9	10251	VICTE	3	7/8/1996, 12:00:00 AM	8/5/1996, 12:00:00 AM	7/15/1996, 12:00:00 AM
10	10251	VICTE	3	7/8/1996, 12:00:00 AM	8/5/1996, 12:00:00 AM	7/15/1996, 12:00:00 AM
11	10251	VICTE	3	7/8/1996, 12:00:00 AM	8/5/1996, 12:00:00 AM	7/15/1996, 12:00:00 AM
12	10252	SUPRD	4	7/9/1996, 12:00:00 AM	8/6/1996, 12:00:00 AM	7/11/1996, 12:00:00 AM
13	10252	SUPRD	4	7/9/1996, 12:00:00 AM	8/6/1996, 12:00:00 AM	7/11/1996, 12:00:00 AM
14	10252	SUPRD	4	7/9/1996, 12:00:00 AM	8/6/1996, 12:00:00 AM	7/11/1996, 12:00:00 AM
15	10253	HANAR	3	7/10/1996, 12:00:00 AM	7/24/1996, 12:00:00 AM	7/16/1996, 12:00:00 AM
16	10253	HANAR	3	7/10/1996, 12:00:00 AM	7/24/1996, 12:00:00 AM	7/16/1996, 12:00:00 AM
17	10253	HANAR	3	7/10/1996, 12:00:00 AM	7/24/1996, 12:00:00 AM	7/16/1996, 12:00:00 AM
18	10254	CHOPS	5	7/11/1996, 12:00:00 AM	8/8/1996, 12:00:00 AM	7/23/1996, 12:00:00 AM
19	10254	CHOPS	5	7/11/1996, 12:00:00 AM	8/8/1996, 12:00:00 AM	7/23/1996, 12:00:00 AM
20	10254	CHOPS	5	7/11/1996, 12:00:00 AM	8/8/1996, 12:00:00 AM	7/23/1996, 12:00:00 AM
21	10255	RICSU	9	7/12/1996, 12:00:00 AM	8/9/1996, 12:00:00 AM	7/15/1996, 12:00:00 AM
22	10255	RICSU	9	7/12/1996, 12:00:00 AM	8/9/1996, 12:00:00 AM	7/15/1996, 12:00:00 AM

Completed (0.81 s) Columns: 21 Rows: 99+

Query settings: Properties, Name: Orders, Applied steps: Source, Navigation 1, Expanded..., Added custom..., Reordered...

Data destination: No data destination

Step 3- Remove Unnecessary Columns

Unnecessary columns were removed from the Products table to retain only the fields required for further analysis.

Power Query Editor

Home Transform Add column View Help

Close & Apply, Get data, Enter data, Manage parameters, Refresh, Properties, Advanced editor, Export query results, Choose columns, Remove columns, Keep rows, Remove rows, Filter rows, Transform, Combine, Export template

Queries [2]: Orders, Products

Table.TransformColumnTypes(#"Removed other columns", {"UnitsInStock", Int64.Type})

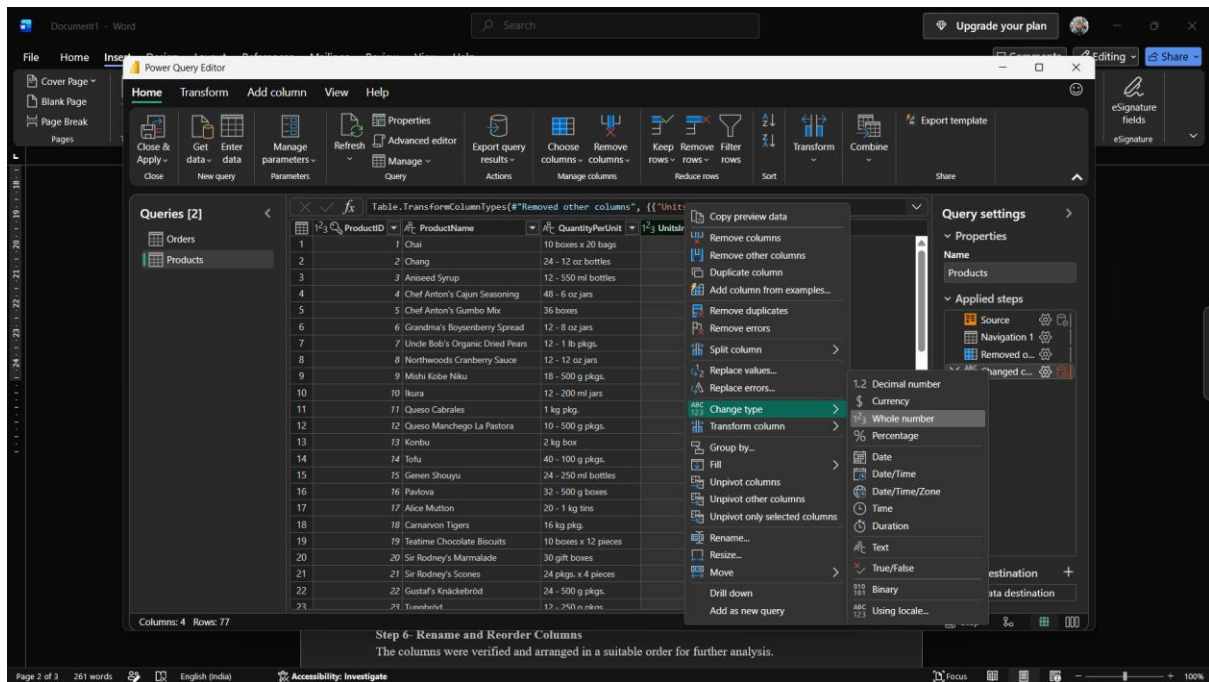
ProductID	ProductName	QuantityPerUnit	UnitsInStock
1	Chai	10 boxes x 20 bags	39
2	Chang	24 - 12 oz bottles	17
3	Aniseed Syrup	12 - 550 ml bottles	13
4	Chef Anton's Cajun Seasoning	48 - 6 oz jars	53
5	Chef Anton's Gumbo Mix	36 boxes	0
6	Grandma's Boysenberry Spread	12 - 8 oz jars	120
7	Uncle Bob's Organic Dried Pears	12 - 1 lb pkgs.	15
8	Northwoods Cranberry Sauce	12 - 12 oz jars	6
9	Mishi Kobe Niku	18 - 500 g pkgs.	29
10	Ikura	12 - 200 ml jars	31
11	Queso Cabrales	1 kg pkg.	22
12	Queso Manchego La Pastora	10 - 500 g pkgs.	86
13	Konbu	2 kg box	24
14	Tofu	40 - 100 g pkgs.	35
15	Genen Shouyu	24 - 250 ml bottles	39
16	Pavlova	32 - 500 g boxes	29
17	Alice Mutton	20 - 1 kg tins	0
18	Carnarvon Tigers	16 kg pkg.	42
19	Teatime Chocolate Biscuits	10 boxes x 12 pieces	25
20	Sir Rodney's Marmalade	30 gift boxes	40
21	Sir Rodney's Scones	24 pkgs. x 4 pieces	3
22	Gustaf's Knäckebröd	24 - 500 g pkgs.	104
23	Troutmerry	12 - 250 g pkgs.	61

Query settings: Properties, Name: Products, Applied steps: Source, Navigation 1, Removed other columns, Changed column type...

Data destination: No data destination

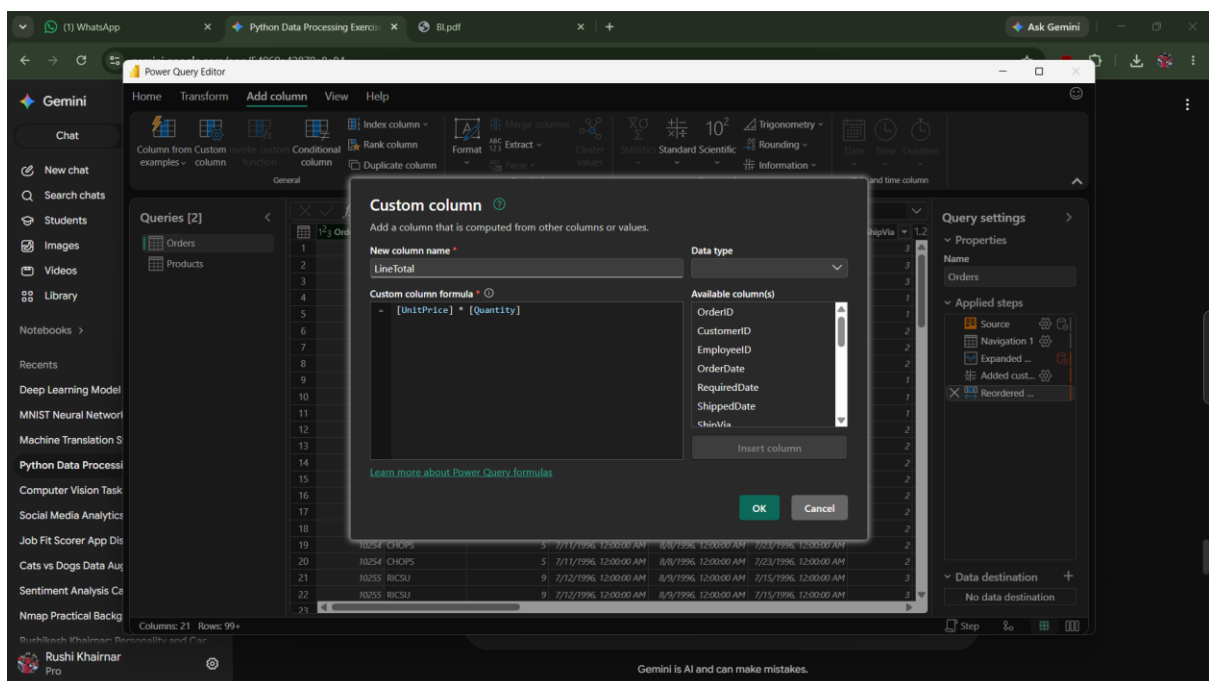
Step 4 - Verify the Data Type of UnitsInStock

The data type of the UnitsInStock column was verified as Whole Number, which is appropriate for representing stock quantities.



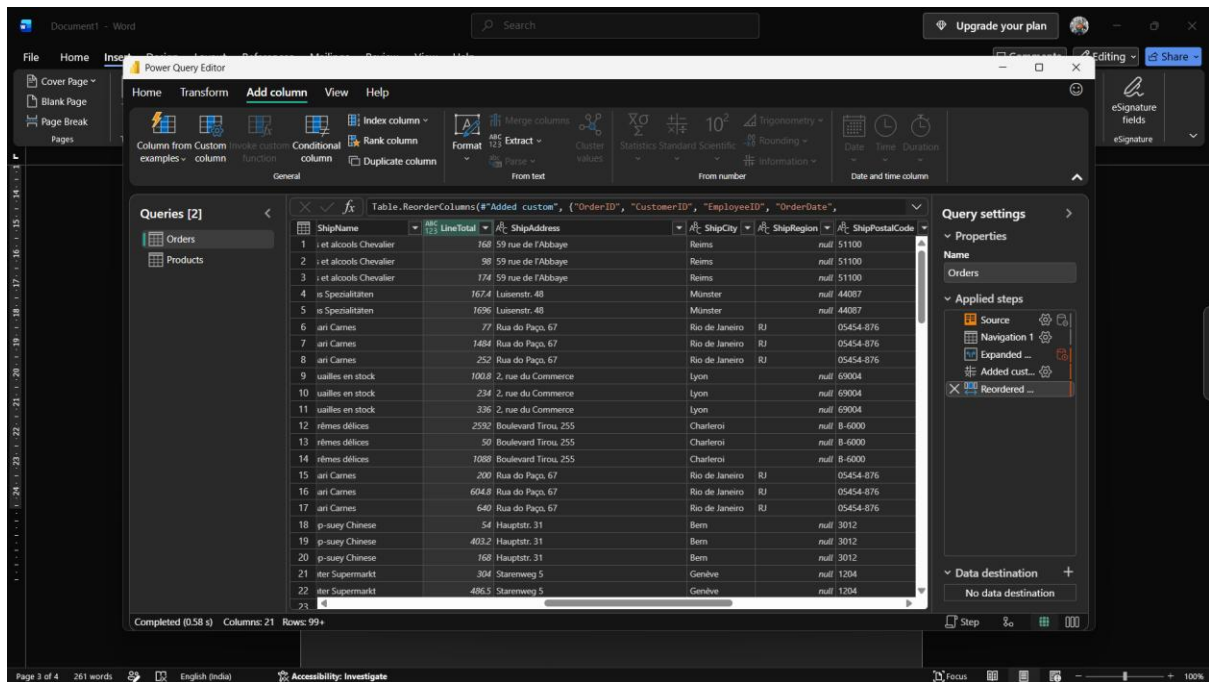
Step 5- Calculate Line Total

A new LineTotal column was created by multiplying UnitPrice by Quantity for each order detail.



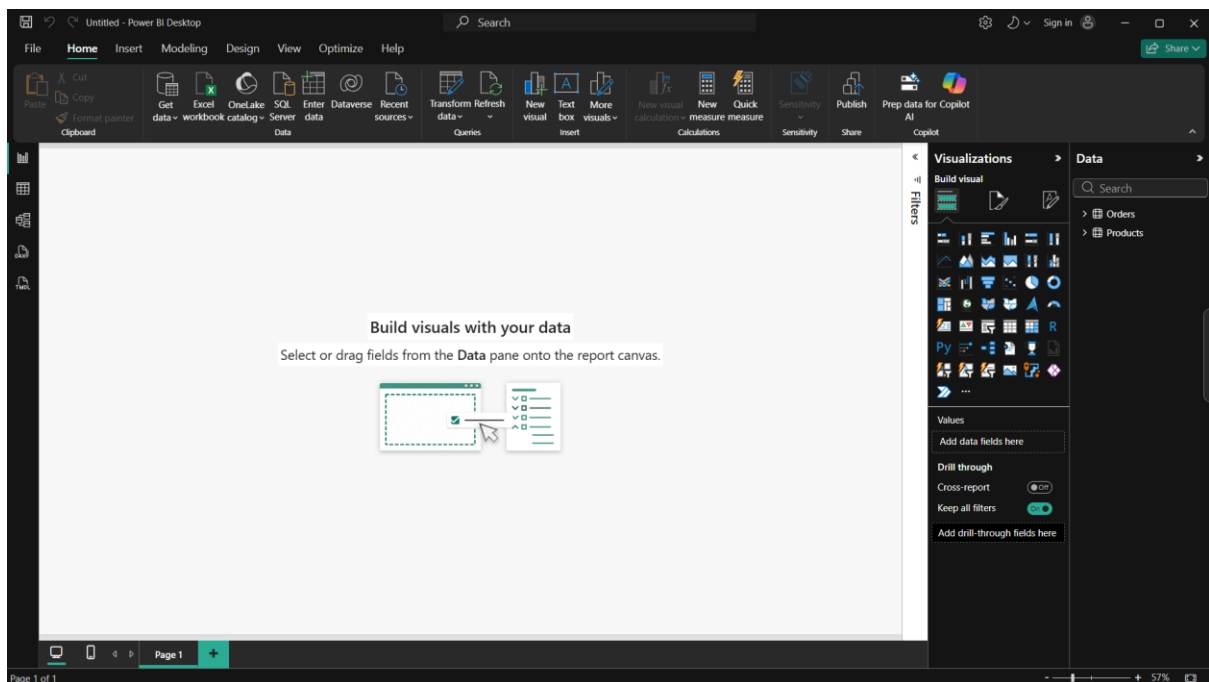
Step 6- Rename and Reorder Columns

The columns were verified and arranged in a suitable order for further analysis.



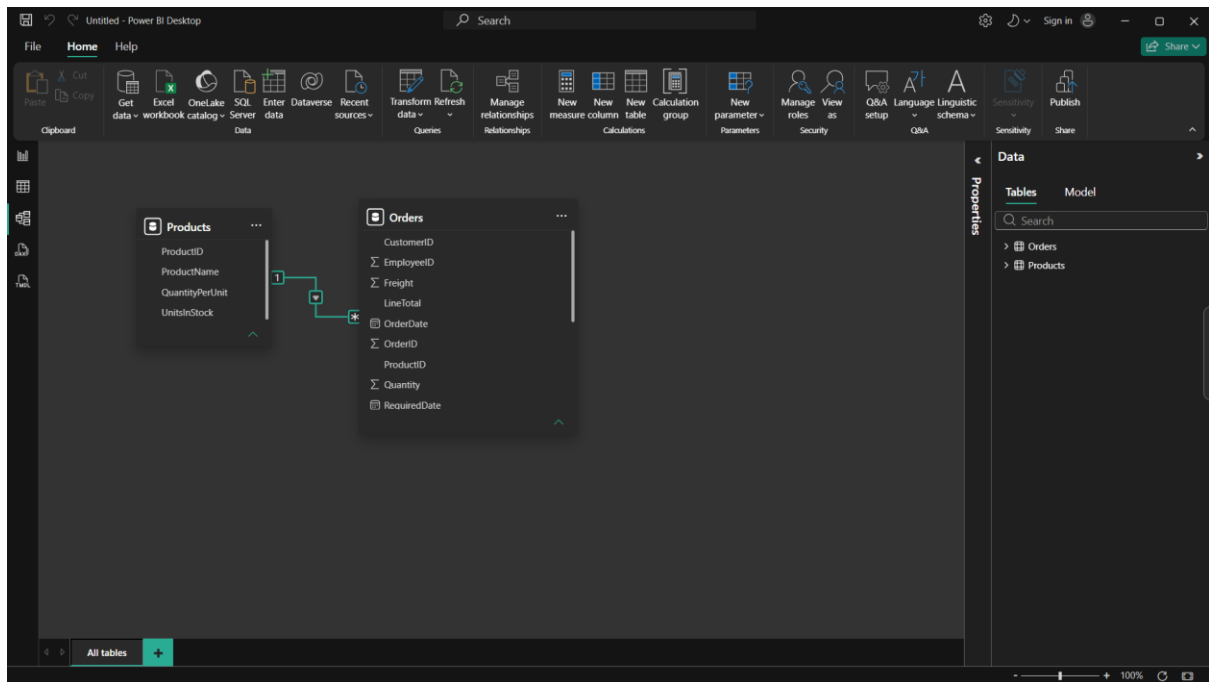
Step 7- Load the Transformed Data

The transformed data was loaded into Power BI Desktop using the Close & Apply option.



Step 8 Create Relationships Between Tables

Relationships were created between the related tables using their common key fields to establish the data model.



Step 9 Result

The ETL process was successfully performed in Power BI by extracting data from an Excel workbook, transforming the data using Power Query, creating the Line Total column, loading the transformed data, and establishing relationships between the tables.

